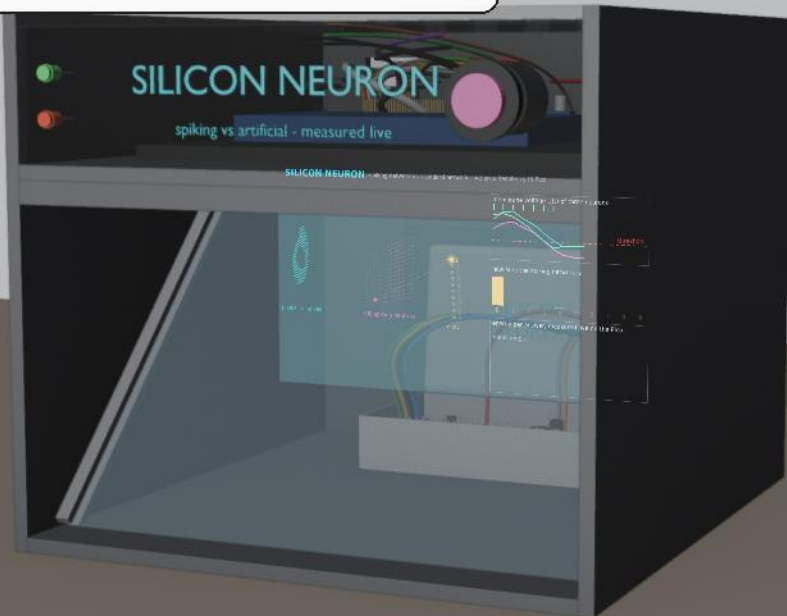


The finished demo

A spiking network and a normal one race on a Raspberry Pi Pico; their spikes and measured energy float in mid-air.



1

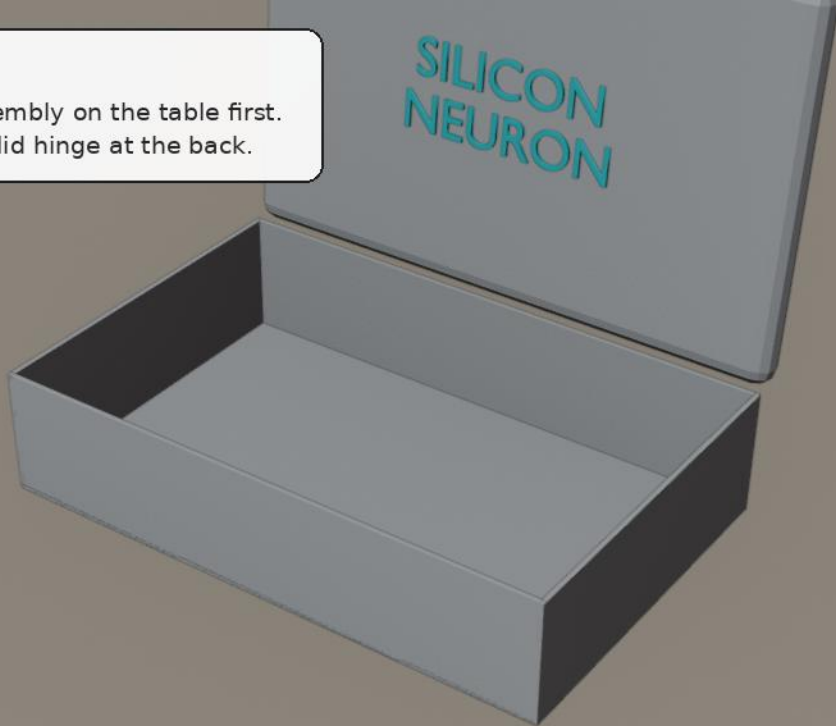
What's in the kit

Every part, exploded. The number on each part is the step that uses it.



2**Mint tin**

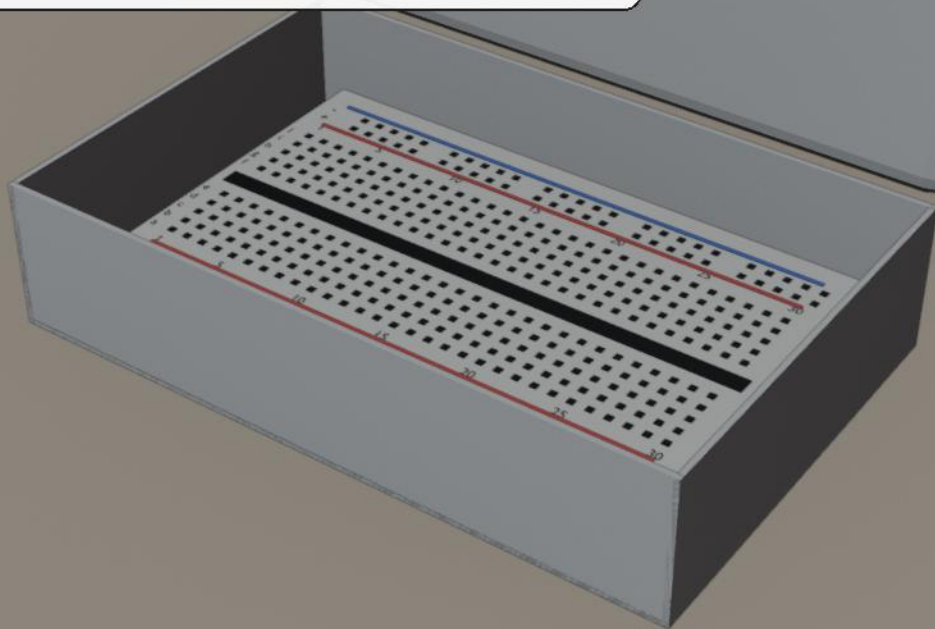
Build the Pico sub-assembly on the table first.
Empty mint tin, open, lid hinge at the back.



3

Half-size breadboard

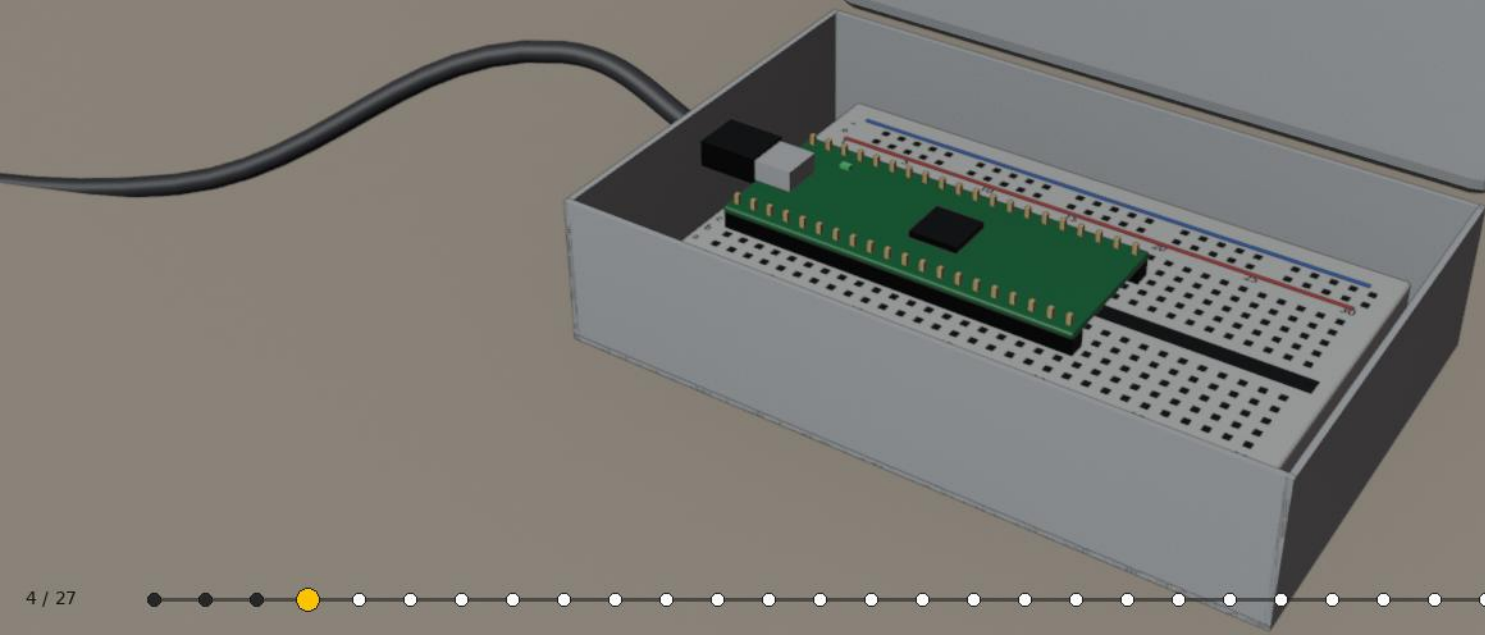
Peel the backing and stick the 400-point breadboard into the tin, rows running left-right, red (+) rail at the BACK.



4

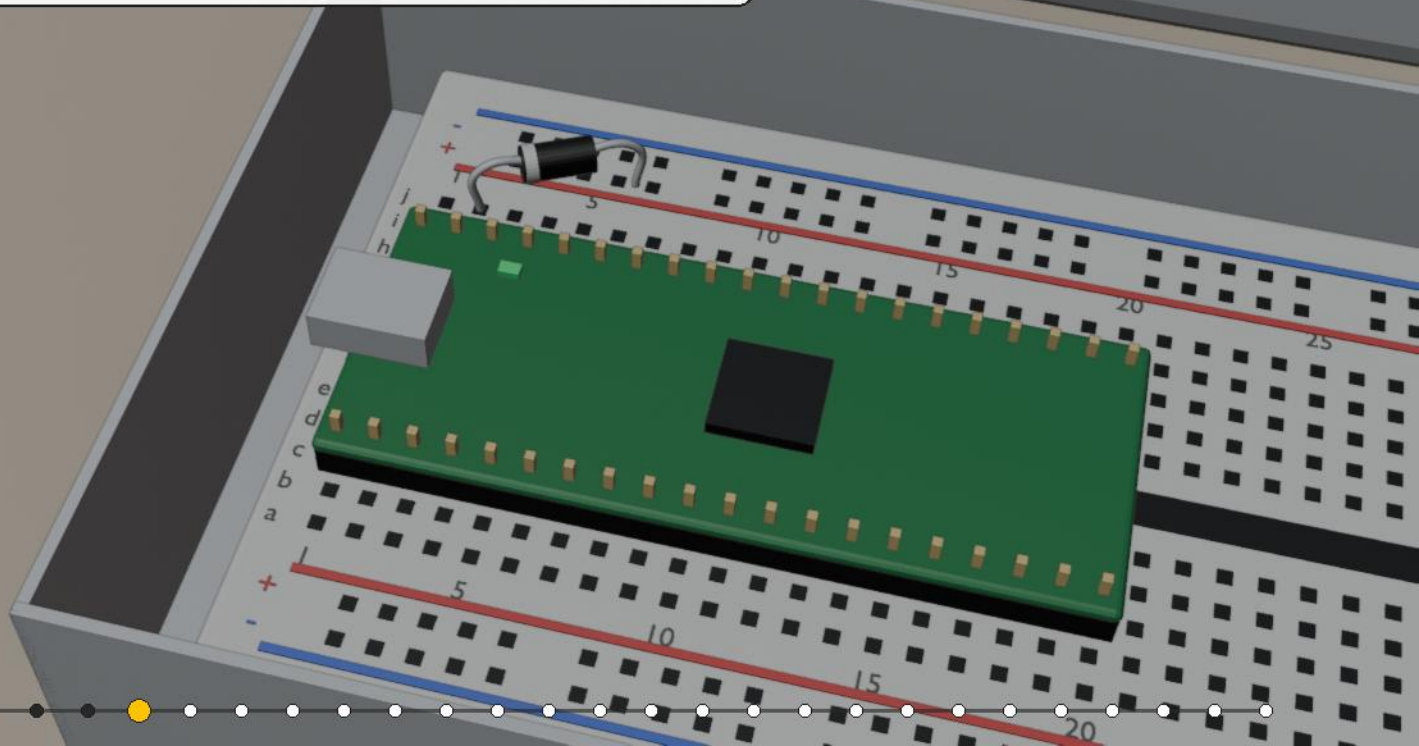
Raspberry Pi Pico

Flash it first: hold BOOTSEL, plug in the micro-USB to USB-A cable, drag the .uf2 on, unplug. Press it into rows 1-20, USB end LEFT, pins in columns c and h.



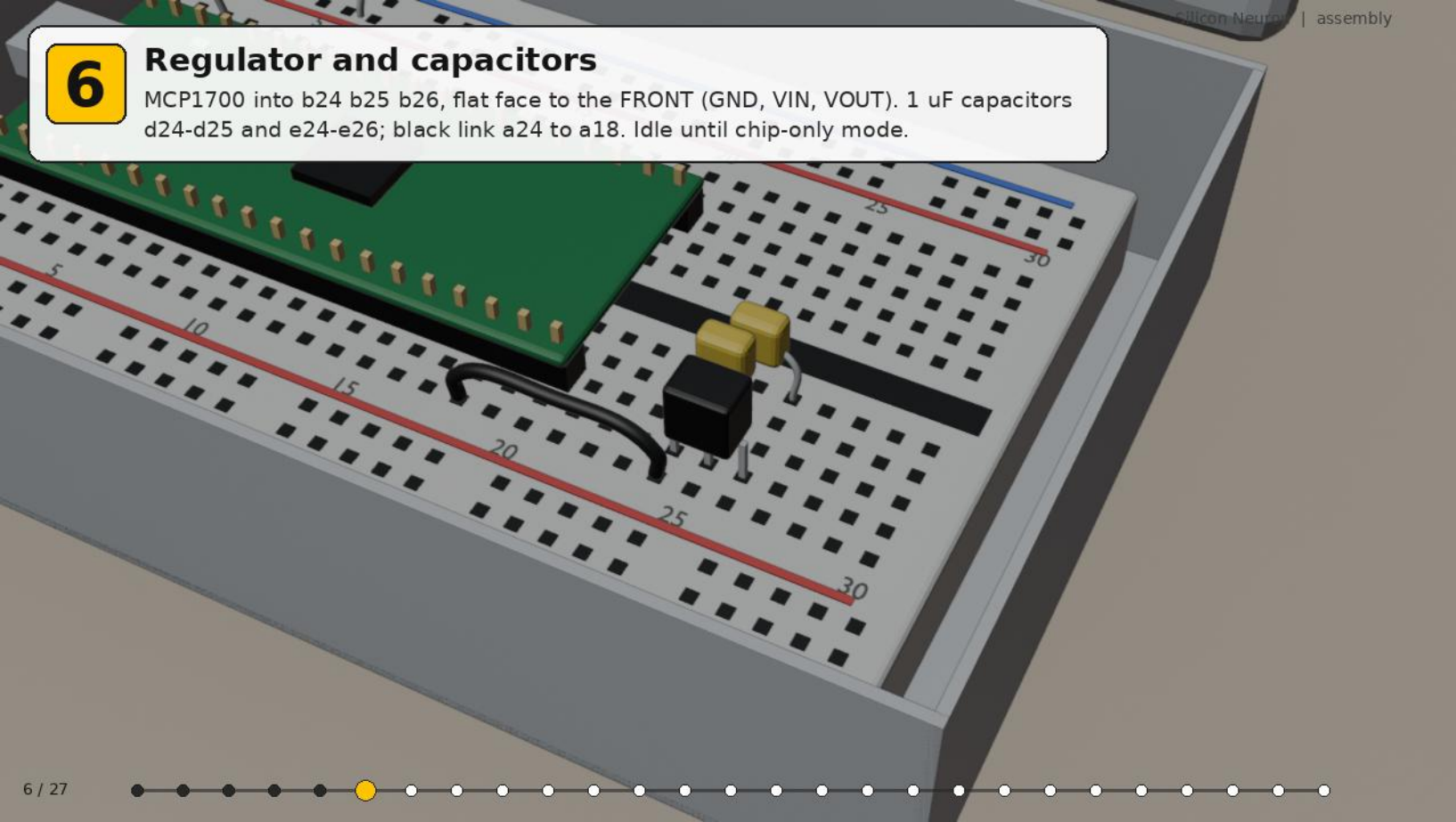
5**Schottky diode 1N5819**

Plain end (anode) into the back red + rail at row 6; banded end (cathode) into j2, the strip of Pico pin 39, VSYS.



6 Regulator and capacitors

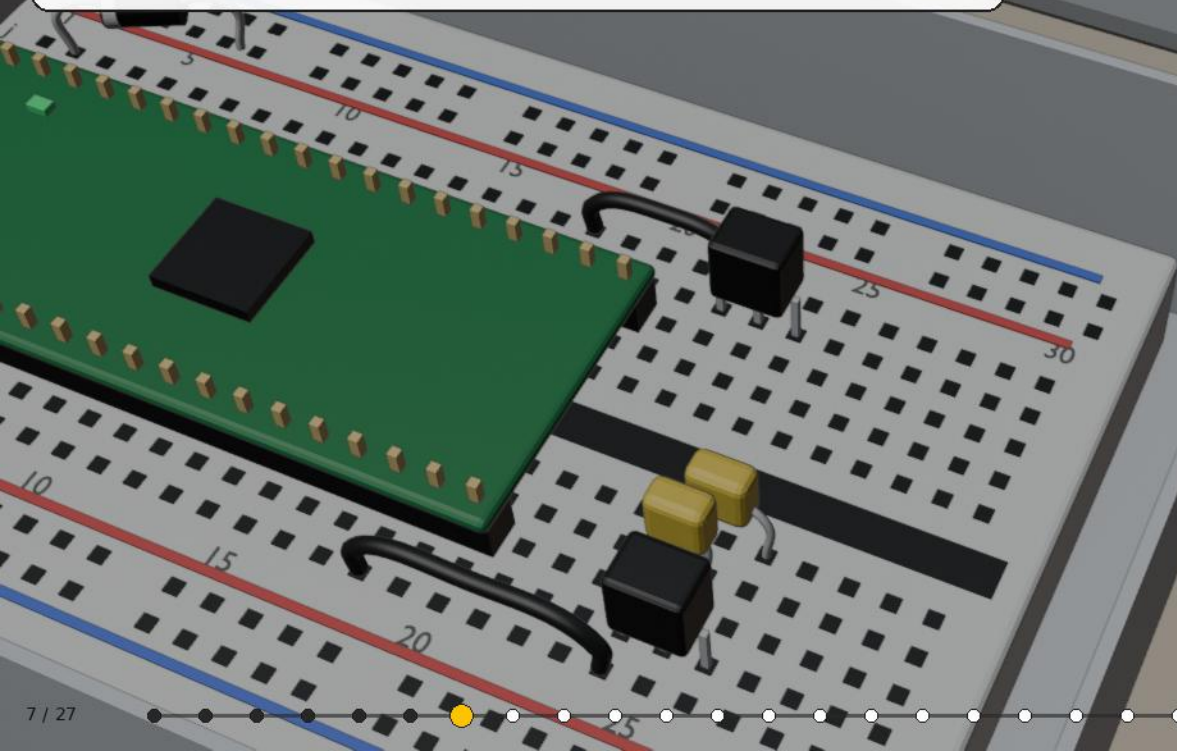
MCP1700 into b24 b25 b26, flat face to the FRONT (GND, VIN, VOUT). 1 μ F capacitors d24-d25 and e24-e26; black link a24 to a18. Idle until chip-only mode.



7

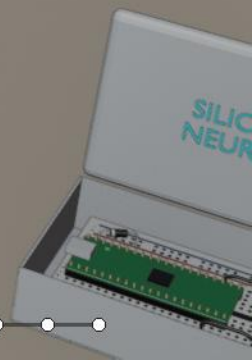
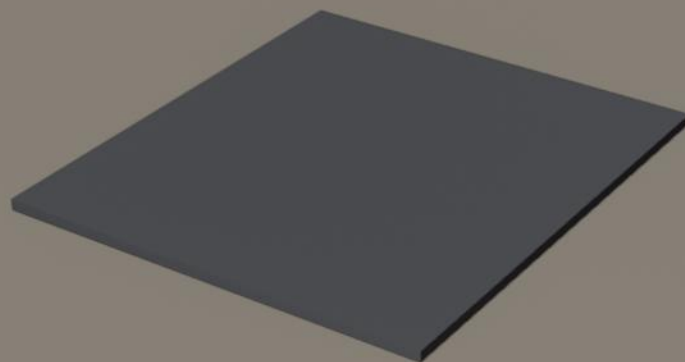
Temperature probe

DS18B20 into i22 i23 i24, flat face to the FRONT (GND, DQ, VDD);
black link j22 to j18 (Pico GND). It logs the chip's temperature.



8**Floor**

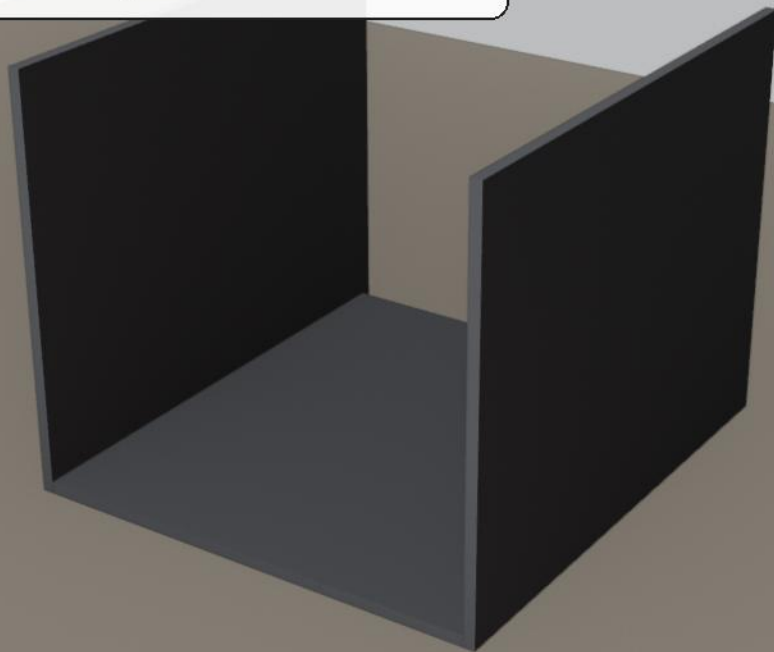
Black foam board, 196 x 220 mm, flat on the table.



9

Side walls

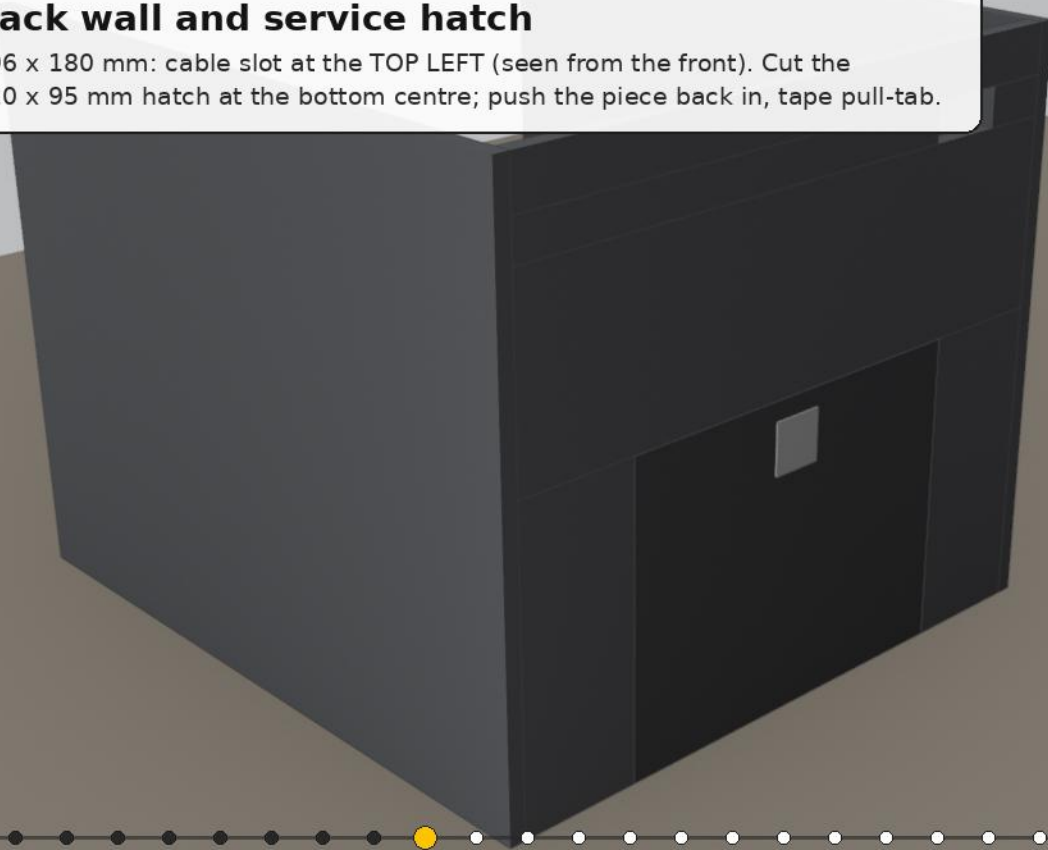
Hot-glue both 225 x 180 mm sides against the floor edges.
Check the corners with a set square.



10

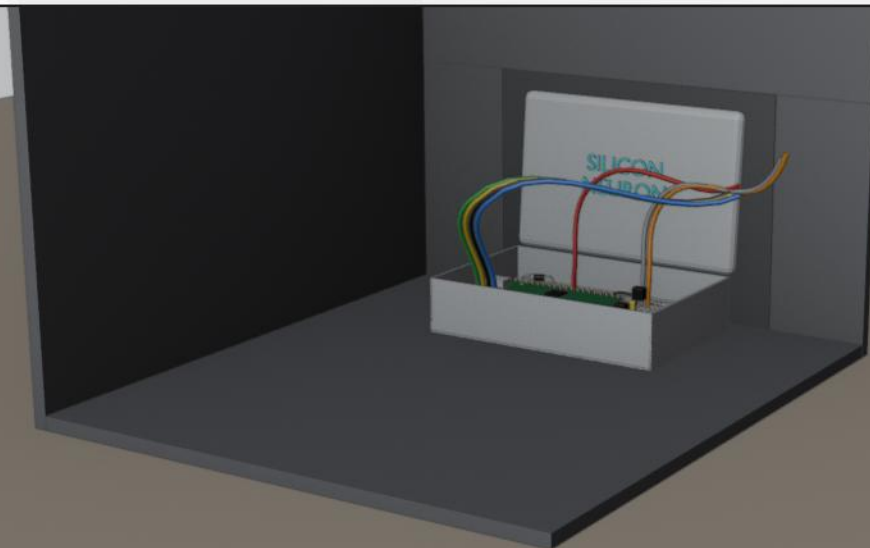
Back wall and service hatch

196 x 180 mm: cable slot at the TOP LEFT (seen from the front). Cut the 120 x 95 mm hatch at the bottom centre; push the piece back in, tape pull-tab.



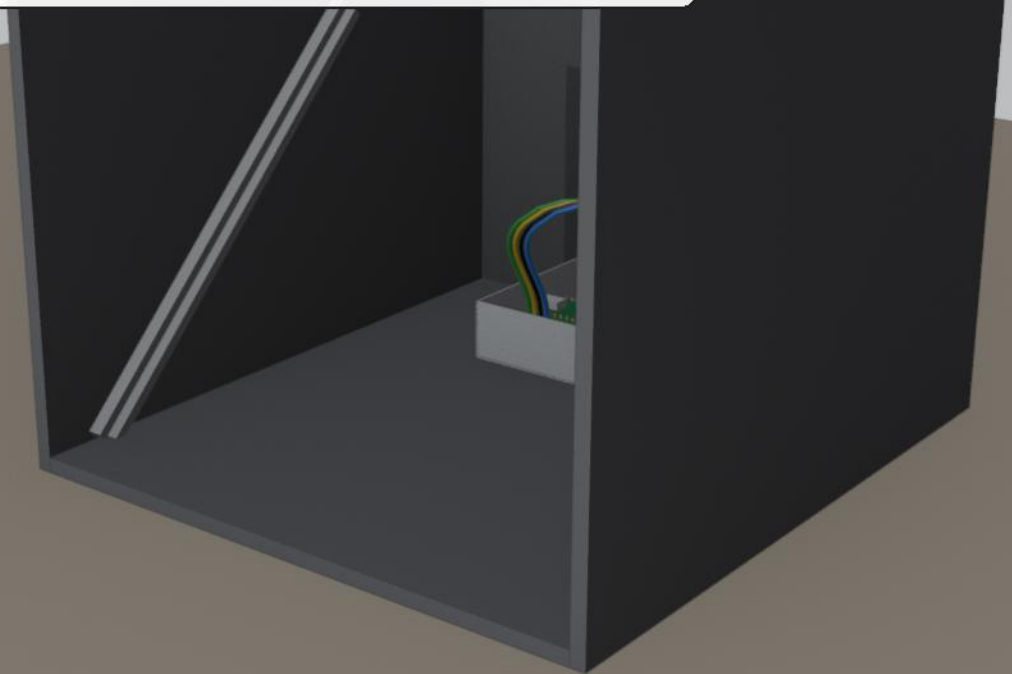
11**Place the Pico tin**

Tin front edge 150 mm behind the opening, centred. Plug in the seven jumpers: red + rail, black a3, yellow a2, green a1, blue a4, grey j23, orange j24. Loose ends back right.



12 Pane rails

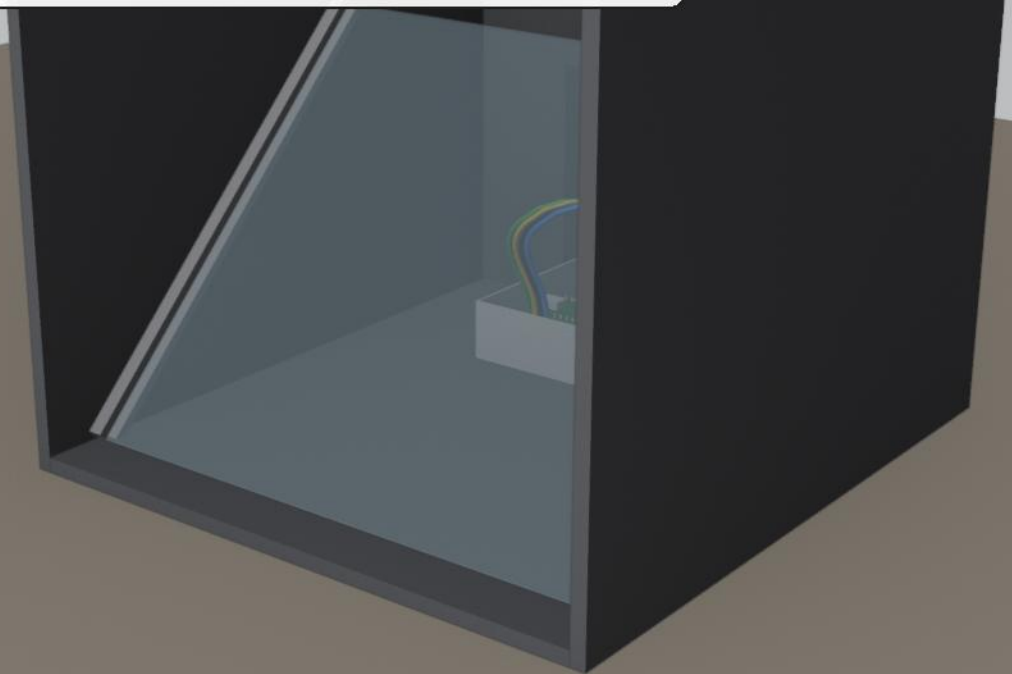
On each side wall draw a 45 degree line from the floor, 20 mm behind the front, up to 125 mm. Glue two rails along it, 3 mm apart.



13

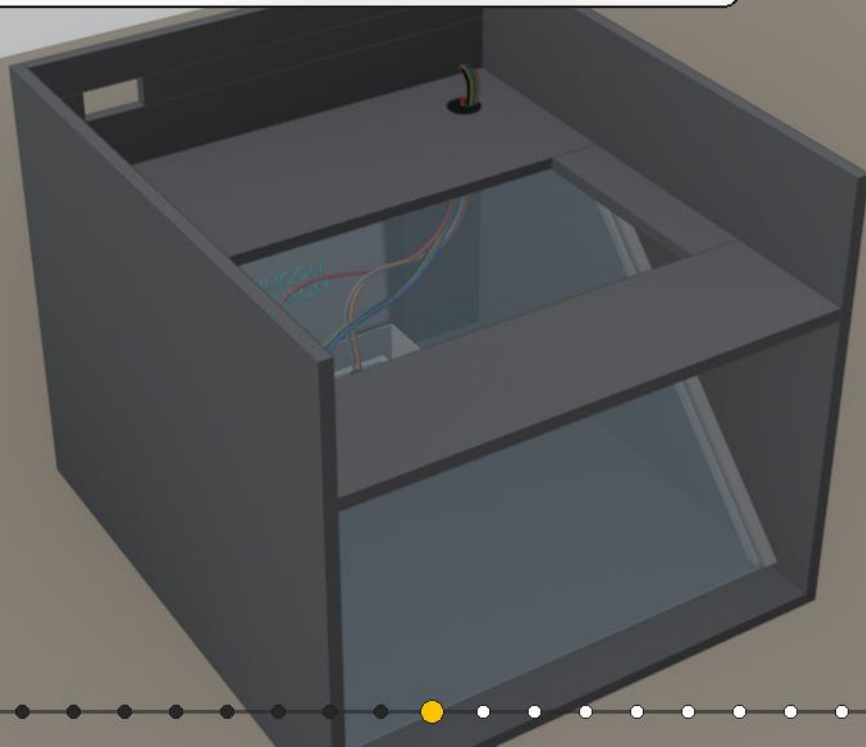
Acrylic pane

Slide the 190 x 170 mm, 2 mm sheet down the slots, film still on.
It touches the floor at the front and reaches 140 mm back at the top.



14 Roof

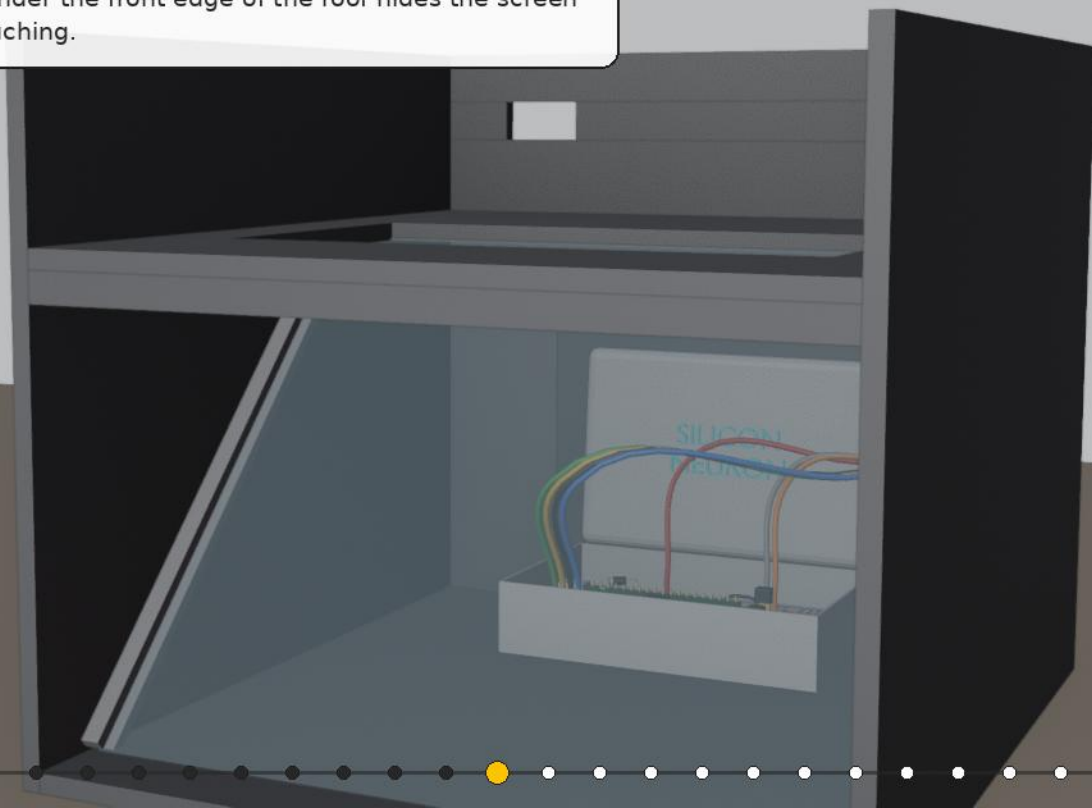
Glue the roof 125 mm above the table, screen window towards the back.
Feed the seven loose jumpers up through the 14 mm hole at the back right.



15

Valance

The 8 mm strip under the front edge of the roof hides the screen from anyone crouching.



16**Screen, face down**

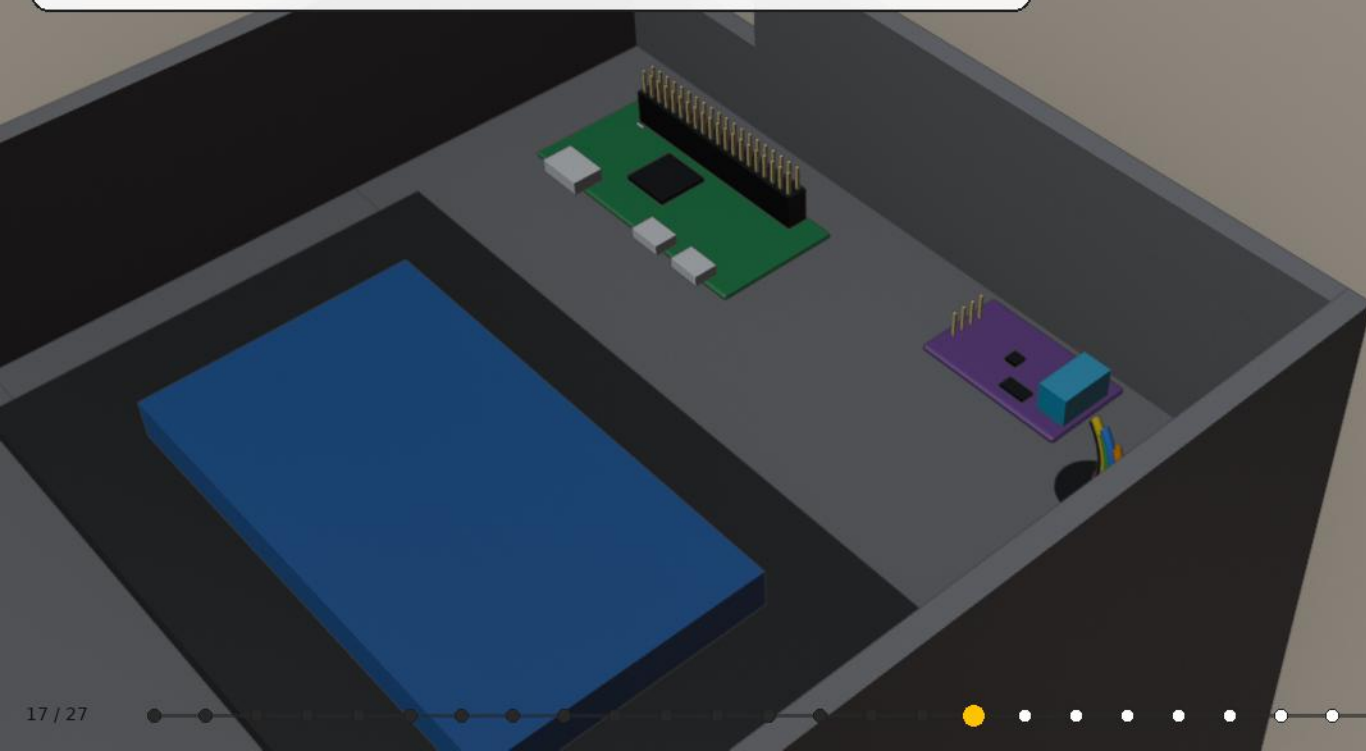
Lay the 7-inch screen FACE DOWN over the window, its lit area's back edge right above the top of the pane. Tape the bezel.



17

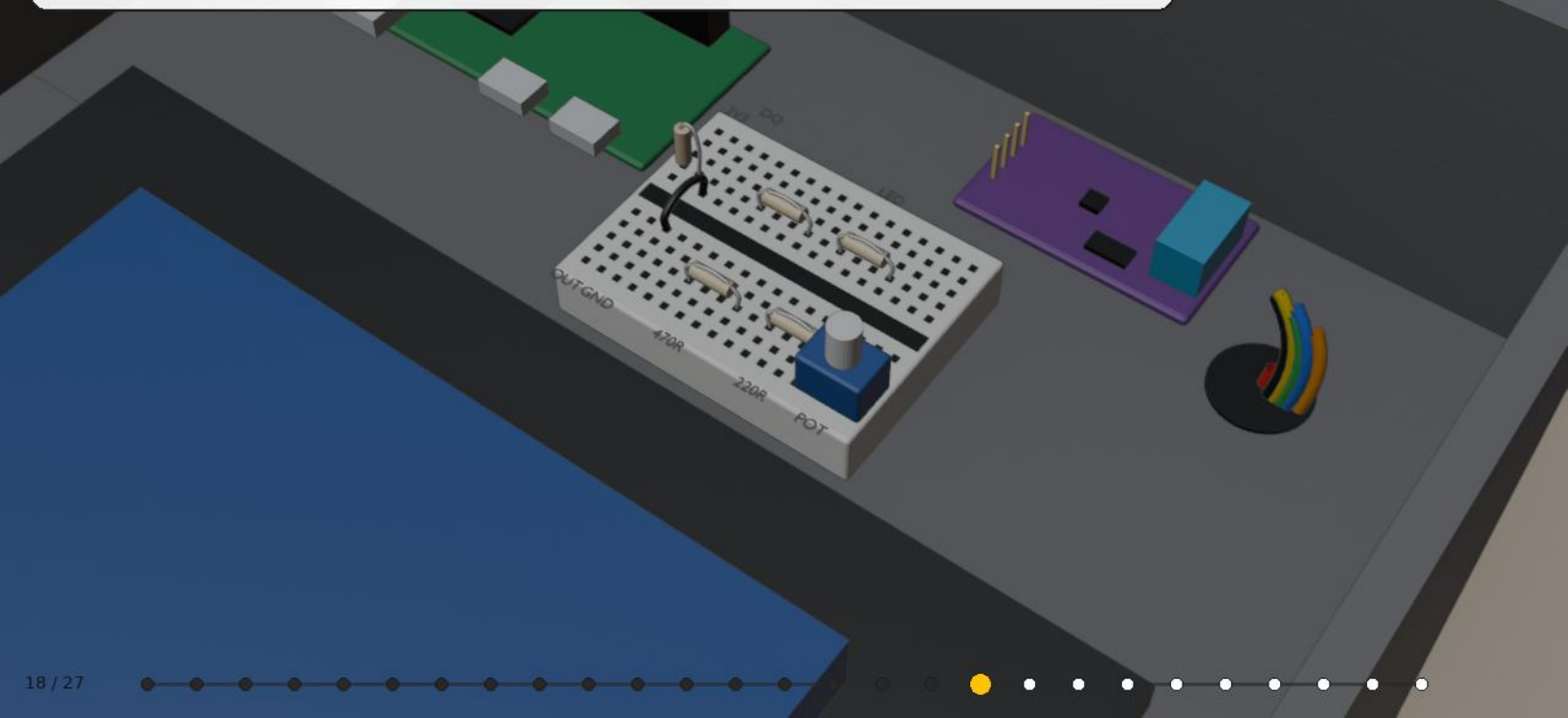
Pi Zero 2 W and INA226

Foam-tape both to the roof behind the screen: Pi on the left (header at the back), the meter along the back edge on the right.



18 Calibration board

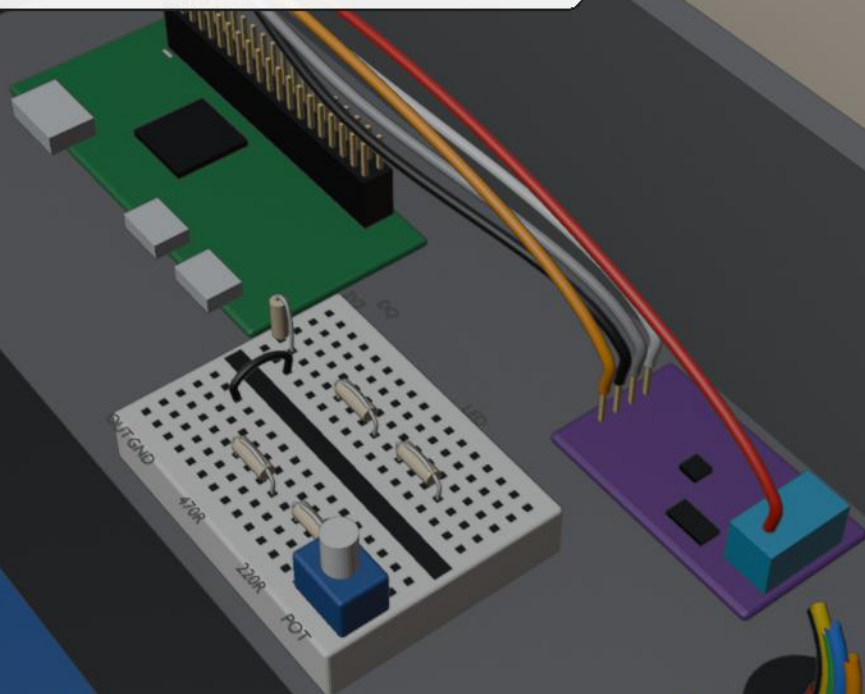
Mini breadboard in front of the meter. 4.7k pull-up g1-g2, 330 R g6-g9 and g11-g14, black link e3-f3. Park the 470 R (c6-c9), 220 R (c11-c14) and the pot (b15-b17).



19

Meter wires

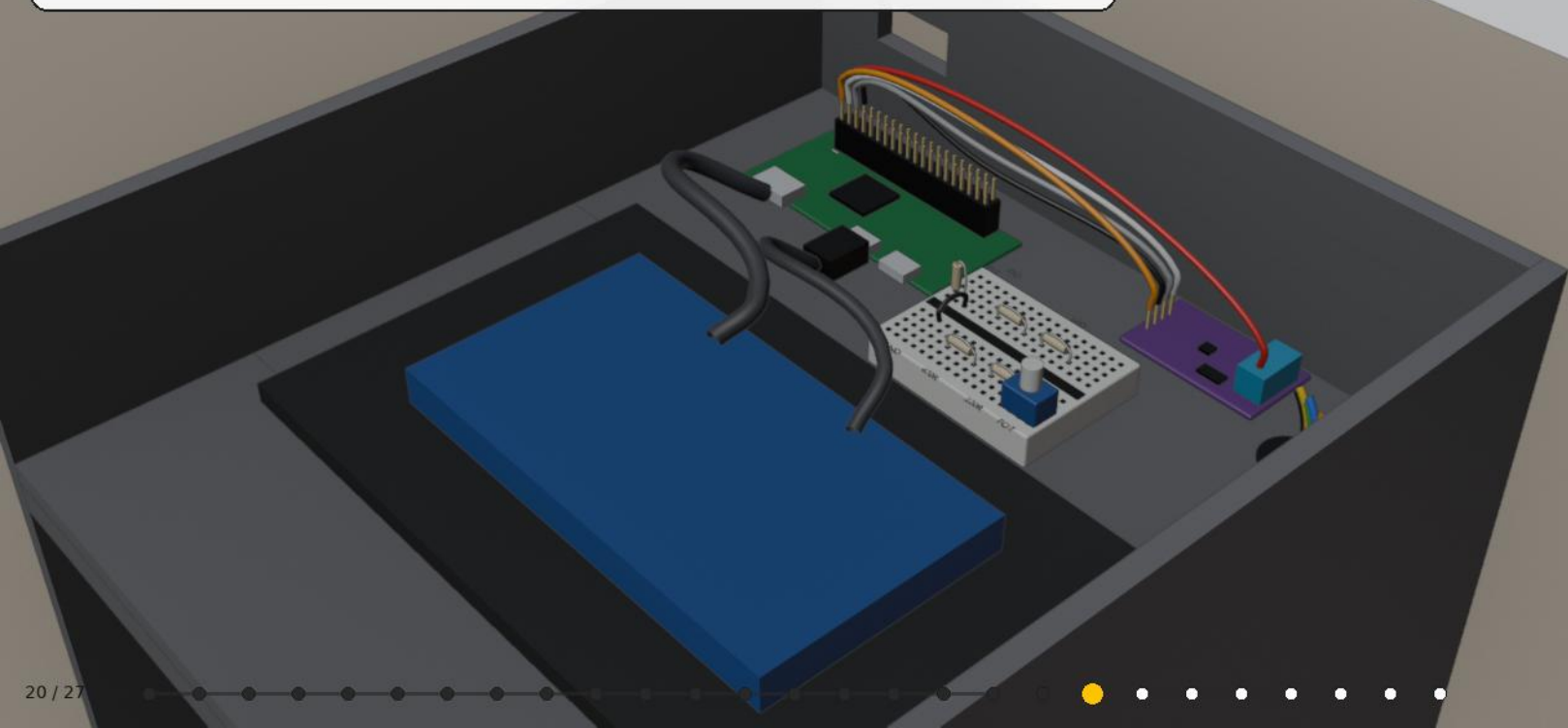
Pi pin 2 (5 V) to VIN+ (red), pin 1 (3.3 V) to VCC (orange), pin 3 to SDA (white), pin 5 to SCL (grey), pin 6 to GND (black).



20

Screen cables

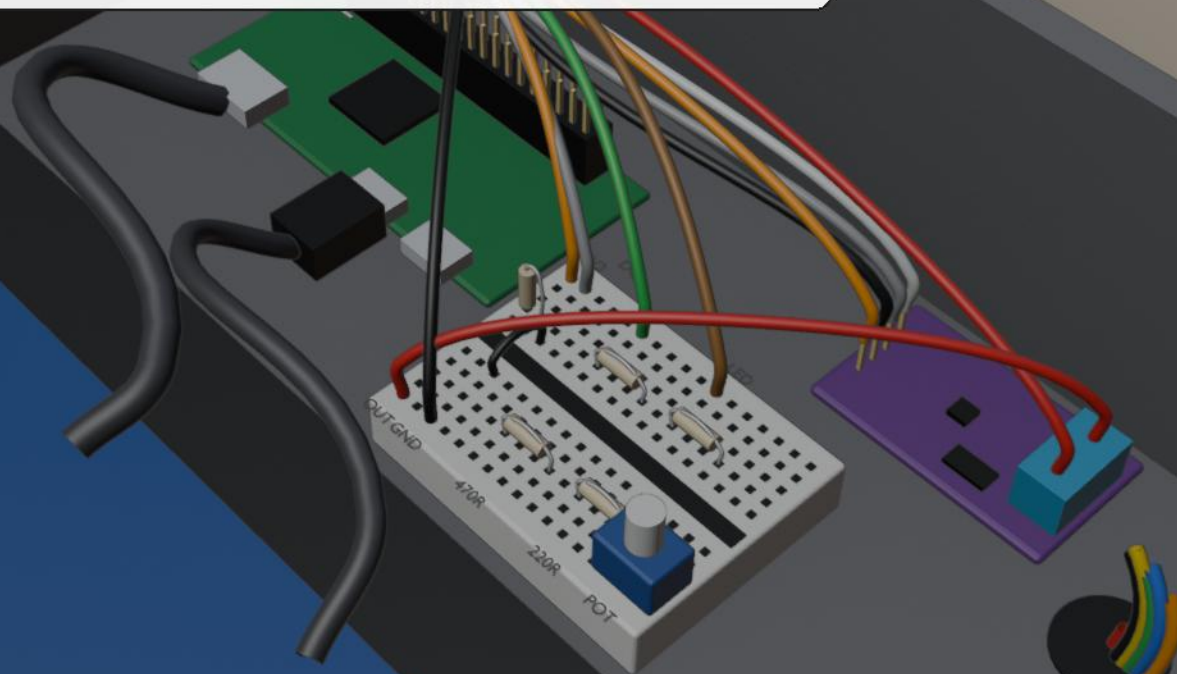
Mini-HDMI from the Pi to the screen. OTG adapter in the Pi's middle port, then the micro-USB to USB-A cable to the screen's USB (power and touch).



21

Calibration board wires

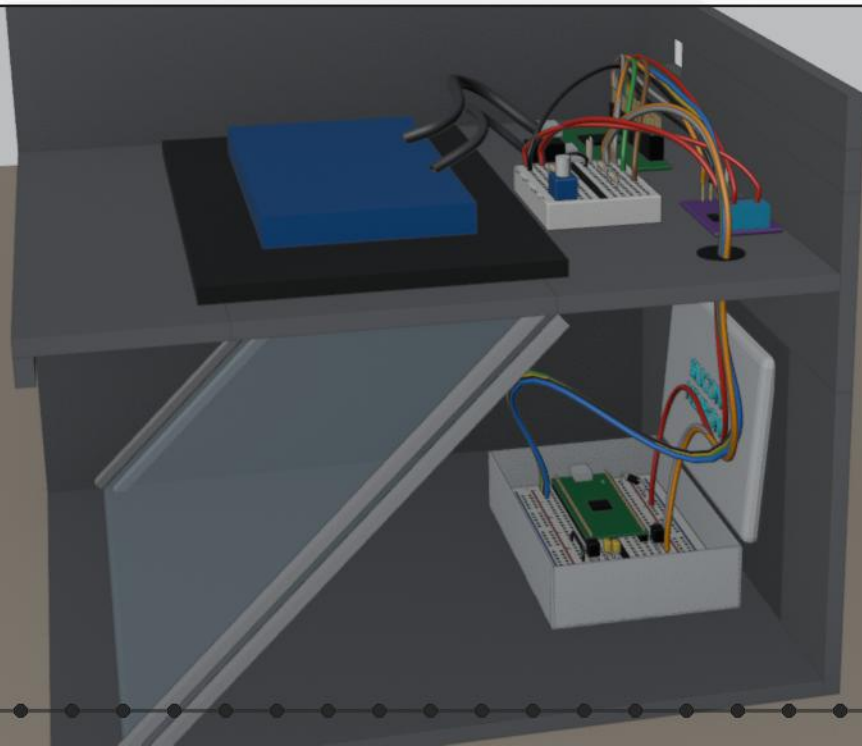
Short red: meter VIN- to OUT (a1). Pi pin 20 to a3 (black), 17 to j1 (orange), 7 to j2 (grey), 15 to j6 (green), 16 to j11 (brown).



22

Pico wires

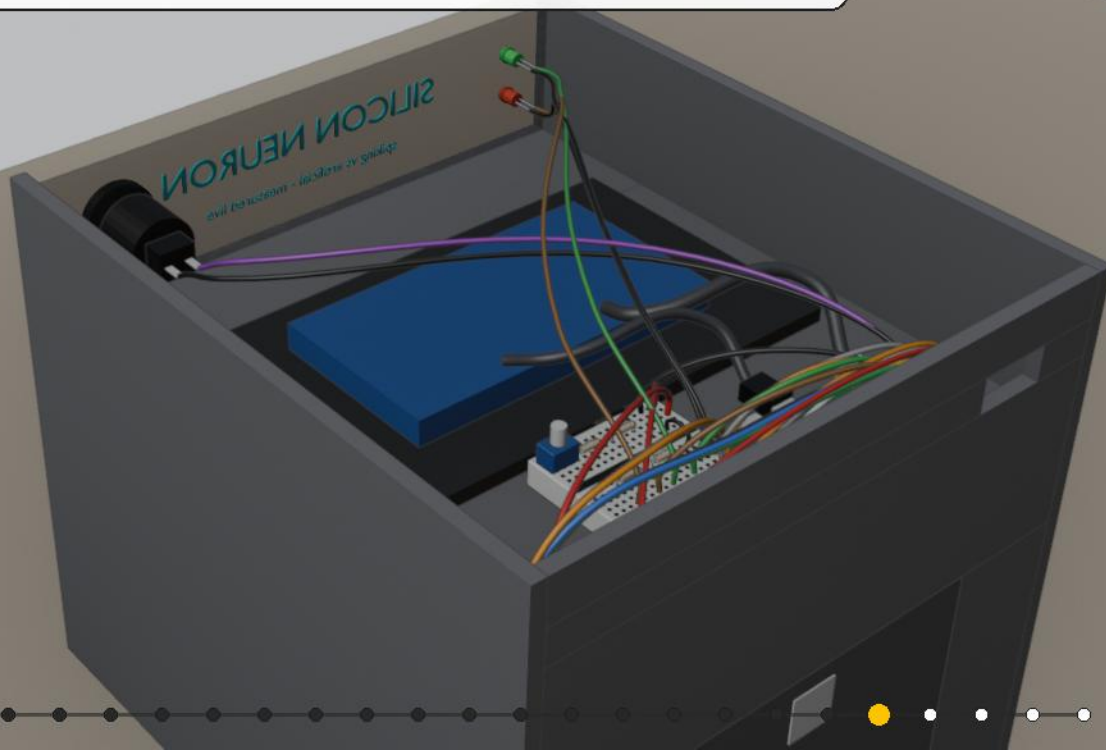
Red into OUT c1 (the meter, never the Pi's 5 V). Grey to i2, orange to i1. Yellow to Pi pin 8, green to 10, blue to 11, black to 9. (Right wall hidden here.)



23

Button, LEDs and cap front

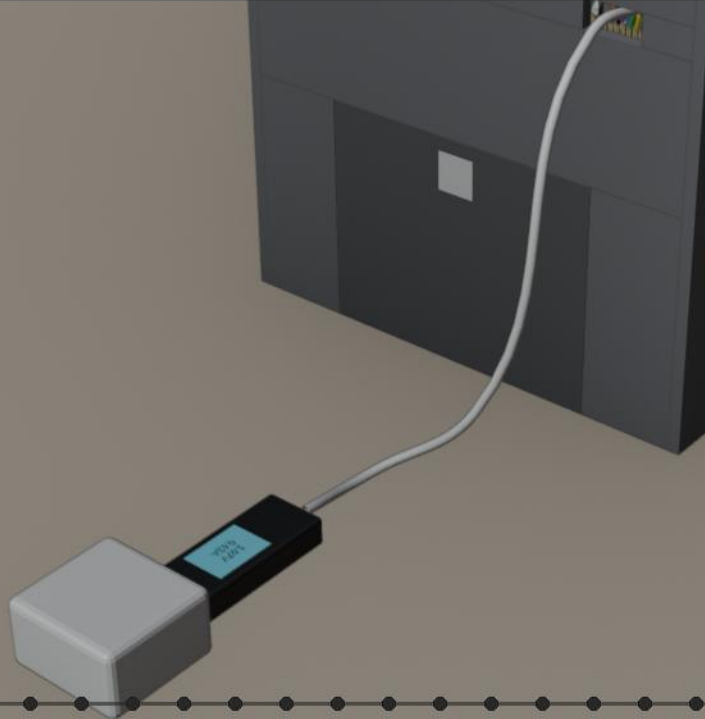
24 mm button and two 5 mm LEDs (long leg = +) into the cap front, glue it on. Purple pin 13 + black 14 to the button; green j9, brown j14 to LED +; black h3, i3 to LED -.



24

Lid, USB meter and power

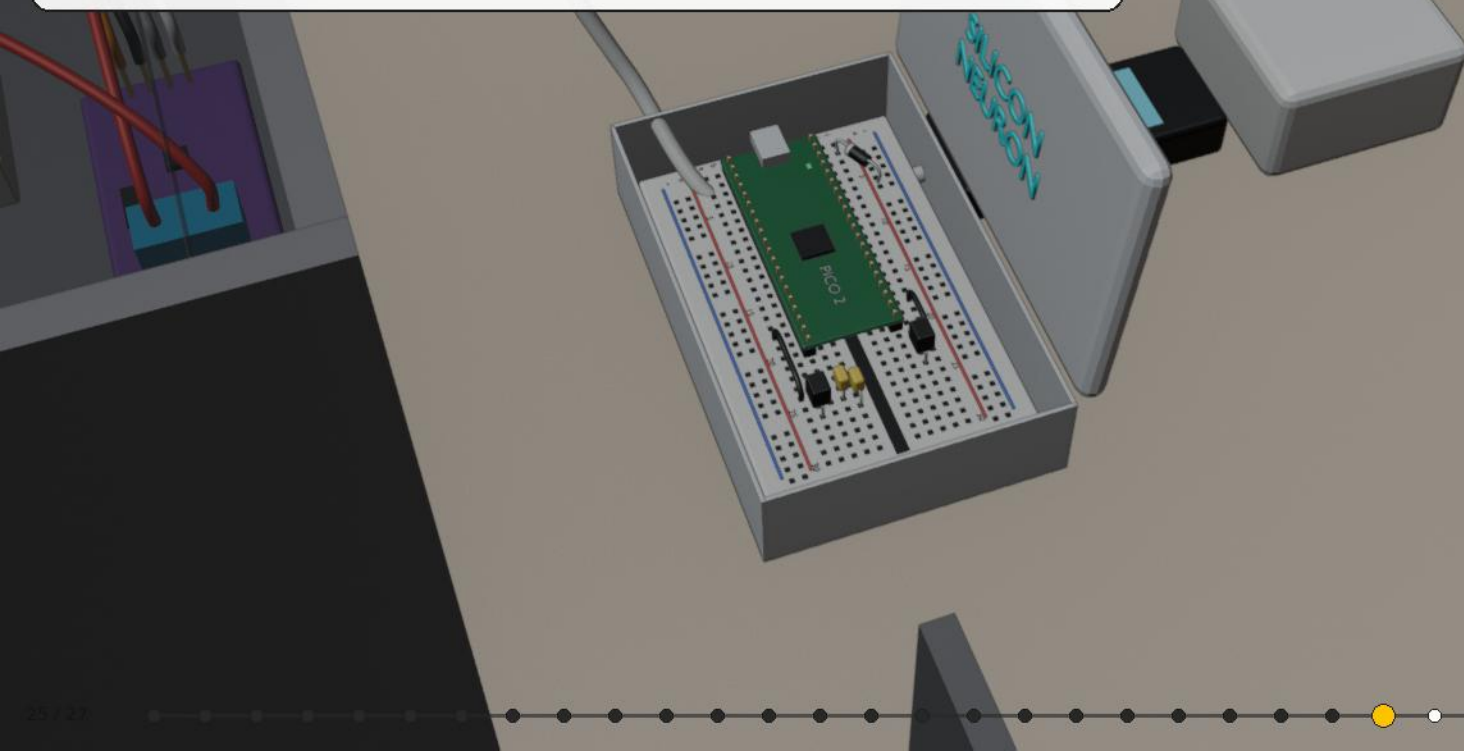
Lid on (unglued). Power lead out through the back slot, through the USB inline meter, into a 5 V USB charger. The meter shows the whole demo's draw.



25

Later: swap to Pico 2

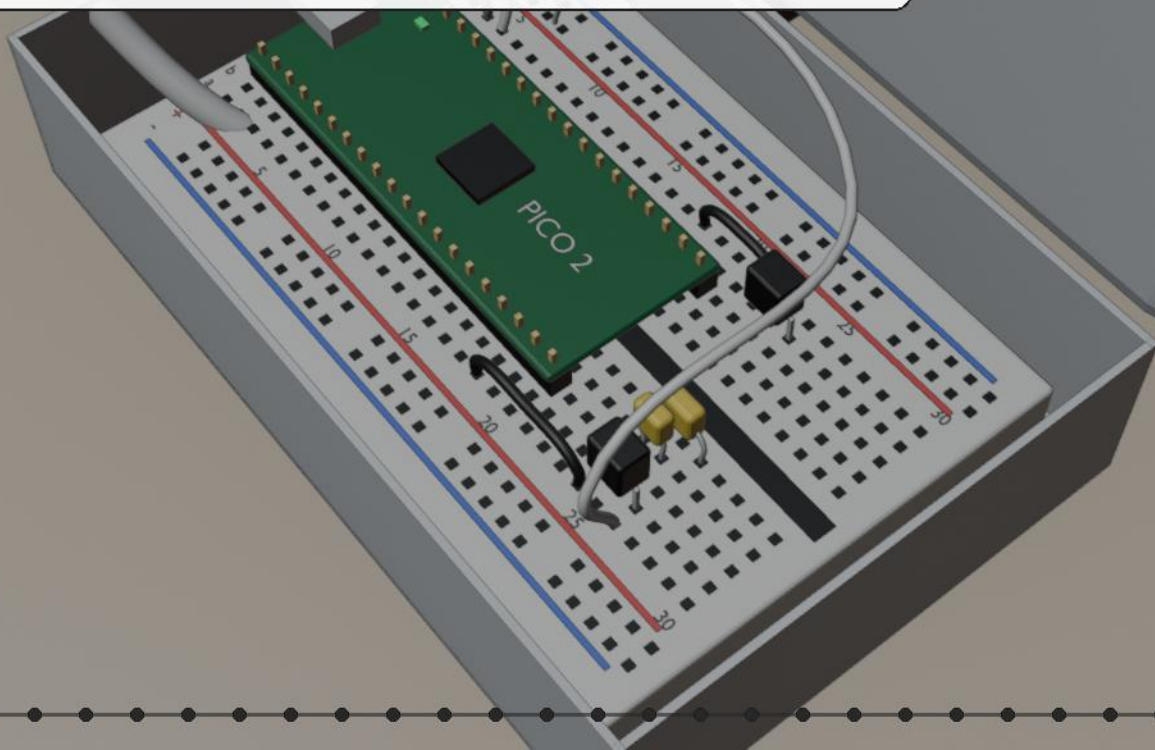
Pull the hatch, slide the tin out (jumpers stay plugged in; hidden here). Lift the Pico out, press the Pico 2 into the same holes. Flash it with its own .uf2.



26

Later: chip-only power mode

Red jumper from the + rail to a25 (regulator VIN), white link a26 to j5 (3V3), grey link j4 to j3 (3V3_EN to GND). Run bench.py --rail 3v3.



27

Power on

Slide the tin back, hatch in. Dim the lights: the display floats in front of the Pico that computes it. Press the button for the next digit.

